

$$3d^1$$

June 11, 2026

$$1 \quad \left| 3d_{yz}^\alpha 3d_{xz}^0 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle$$

Here are the CSFs:

$$\begin{aligned} |1, 1/2, 1/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\ |2, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\ |3, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\ |4, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\ |5, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\ |6, 1/2, -1/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\ |7, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\ |8, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\ |9, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\ |10, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \end{aligned}$$

The Ground State CSFs are 2 7

$$\begin{aligned} \left\langle 4, \frac{1}{2}, \frac{1}{2} \left| L_x \right| 2, \frac{1}{2}, \frac{1}{2} \right\rangle &= \sqrt{3}I \\ \left\langle 5, \frac{1}{2}, \frac{1}{2} \left| L_x \right| 2, \frac{1}{2}, \frac{1}{2} \right\rangle &= I \end{aligned}$$

$$\Delta g_{xx}$$

$$-6\zeta\Delta_{2-4}^{-1}$$

$$-2\zeta\Delta_{2-5}^{-1}$$

$$\langle 3, \frac{1}{2}, \frac{1}{2} | L_y | 2, \frac{1}{2}, \frac{1}{2} \rangle = I$$

$$\Delta g_{yy}$$

$$-2\zeta\Delta_{2-3}^{-1}$$

$$\langle 1, \frac{1}{2}, \frac{1}{2} | L_z | 2, \frac{1}{2}, \frac{1}{2} \rangle = -I$$

$$\Delta g_{zz}$$

$$-2\zeta\Delta_{2-1}^{-1}$$